2028 Regatta Venue Wind Analysis

Mid-September Racing Window — Candidate Michigan & Great Lakes Locations

Prepared: June 2026
Analysis window: September 8–14, 2015–2025 (11 years, 2014 excluded — predates HRRR archive)
Racing hours: 10 AM–4 PM EDT (7 hours/day)
Observations per venue: ~523 hourly samples (476 for HYC — see Data Notes)

1. Summary Comparison

All nine candidate venues ranked by mean wind speed during racing hours.

#	Club	Abbreviation	Body of Water	Mean (kt)	Median (kt)	Max (kt)	90th pct (kt)	< 5 kt	5–15 kt	≥ 15 kt	Dom. Dir.	Steadiness R
1	Ludington YC	LYC	Lake Michigan	**8.9**	8.4	22.6	15.3	21%	67%	**12%**	218° (SSW)	0.12
2	Muskegon YC	MYC	Lake Michigan	**8.9**	8.4	21.1	14.8	**18%**	73%	9%	188° (S)	0.18
3	Tawas Bay YC	TBYC	Lake Huron	8.4	8.1	17.3	13.5	19%	**77%**	5%	137° (SE)	0.16
4	Huron YC (Caseville)	HYC	Saginaw Bay	8.3	7.9	19.0	14.3	26%	67%	7%	302° (WNW)	0.20
5	Catawba Island Club	CATYC	Lake Erie	7.7	7.4	17.8	12.5	25%	72%	3%	5° (N)	0.19
6	North Cape YC	NCYC	Lake Erie	7.3	6.7	**22.1**	12.0	27%	70%	3%	69° (ENE)	0.24
7	Crescent Sail YC	CSYC	Lake St. Clair	7.2	6.4	20.1	12.2	30%	67%	3%	47° (NE)	0.27
8	Bayview YC	BYC	Lake St. Clair	6.7	6.3	18.0	11.3	32%	65%	2%	36° (NNE)	0.24
9	Grand Traverse YC	GTYC	Grand Traverse Bay	6.4	6.2	16.4	10.6	**37%**	63%	**1%**	27° (NNE)	**0.04**

> **Steadiness R** is the circular mean resultant length: 0 = direction completely variable, 1 = perfectly steady. All venues are in the 0.04–0.27 range, indicating variable direction is universal across this window — no venue has a "locked-in" breeze.

2. Racing Window Quality

The most useful single metric for race committee planning is the percentage of hours landing in the **5–15 kt racing sweet spot** — enough wind to race confidently without being over-powered for a mixed fleet.

Rank	Club	5–15 kt	< 5 kt	≥ 15 kt	Assessment
1	**TBYC** Tawas Bay	**77%**	19%	5%	Best distribution — strong sweet spot, low heavy-air risk

Rank	Club	5–15 kt	< 5 kt	≥ 15 kt	Assessment
2	**MYC** Muskegon	73%	**18%**	9%	Excellent — lowest drifter rate, manageable heavy air
3	**CATYC** Catawba Island	72%	25%	3%	Good distribution; lighter overall than Lake MI venues
4	**NCYC** North Cape	70%	27%	3%	Solid Lake Erie; watch the light-air days
5	**HYC** Caseville	67%	26%	7%	Saginaw Bay thermal; good when it fills in
6	**LYC** Ludington	67%	21%	**12%**	Windiest venue but highest heavy-air exposure
7	**CSYC** Crescent Sail	67%	30%	3%	Moderate distribution; sheltered location shows
8	**BYC** Bayview	65%	32%	2%	Light end of the fleet; protected St. Clair approach
9	**GTYC** Grand Traverse	63%	**37%**	1%	Highest drifter rate; enclosed bay limits sea breeze

3. Venue Highlights

Lake Michigan — Ludington YC (LYC)

Mean 8.9 kt | 67% sweet spot | 12% heavy air

The most powerful wind resource in the candidate set, with the 90th percentile reaching 15.3 kt — the only venue where double-digit 90th percentile is the norm rather than the exception. The SSW dominant direction (218°) is consistent with the prevailing Lake Michigan sea breeze pattern. The tradeoff is the highest heavy-air exposure (12% ≥15 kt), which has schedule and safety implications for a broad-based fleet. Best suited for a fleet that sails well in a range of conditions.

Lake Michigan — Muskegon YC (MYC)

Mean 8.9 kt | 73% sweet spot | 9% heavy air

Statistically a near-twin to Ludington on raw speed (identical mean and median), but with a meaningfully better distribution: 73% in the sweet spot vs. Ludington's 67%, and 9% heavy air vs. 12%. The lowest drifter rate of all nine venues (18% < 5 kt) makes Muskegon the most reliable option for keeping races moving. Southerly dominant direction (188°) reflects classic lake breeze. **The strongest overall case among Lake Michigan venues.**

Lake Huron — Tawas Bay YC (TBYC)

Mean 8.4 kt | 77% sweet spot | 5% heavy air

The standout finding from this analysis. Tawas Bay posts the highest 5–15 kt percentage of any venue (77%) while keeping heavy air unusually low (5% — less than half of Muskegon's). The cap on maximum wind (17.3 kt — the lowest of all nine venues) reflects the sheltering effect of Tawas Point. For a race committee prioritizing consistency and avoiding schedule disruptions, Tawas Bay's distribution is the cleanest in the dataset. The SE dominant direction (137°) is somewhat atypical — worth monitoring across multiple years for individual-season variability.

Saginaw Bay — Huron YC / Caseville (HYC)

Mean 8.3 kt | 67% sweet spot | 7% heavy air

Saginaw Bay delivers solid thermal-driven afternoon breeze (WNW dominant at 302°, consistent with bay-to-land flow in the afternoon). The 26% drifter rate is a notch above Tawas and the Lake Michigan venues, reflecting the bay's tendency to go light before the thermal fills. **Note:** HYC has 476 observations vs. 523 for most venues — six 2015 archive files were unavailable; statistics are directionally reliable but based on a slightly smaller sample.

Lake Erie — Catawba Island Club (CATYC)

Mean 7.7 kt | 72% sweet spot | 3% heavy air

The most favorable heavy-air profile of the higher-speed venues: only 3% ≥15 kt with a 72% sweet spot rate. Lake Erie's shallow, fetch-driven conditions create a reliable breeze without the extreme events seen at the Lake Michigan sites. The northerly dominant direction (5°) is characteristic of the western Lake Erie summer/fall pattern. A strong choice for a fleet that prioritizes consistency over peak breeze.

Lake Erie — North Cape YC (NCYC)

Mean 7.3 kt | 70% sweet spot | 3% heavy air

Similar profile to Catawba Island but with a notably higher maximum (22.1 kt — highest on Lake Erie in this dataset) suggesting more episodic strong events even if the median is moderate. ENE dominant direction (69°) points up the lake. The 27% drifter rate is slightly elevated. Located at the extreme western end of Lake Erie in La Salle Township, MI; logistics and infrastructure would need vetting.

Lake St. Clair — Crescent Sail YC (CSYC)

Mean 7.2 kt | 67% sweet spot | 3% heavy air

Lake St. Clair produces modest, relatively benign conditions — low heavy-air risk (3%) but 30% drifter rate. The NE dominant direction (47°) is the prevailing fetch direction across the lake. CSYC is a well-established racing club with strong infrastructure, which partially offsets the lighter wind profile. Best considered if operational capacity or travel logistics are primary factors.

Lake St. Clair — Bayview YC (BYC)

Mean 6.7 kt | 65% sweet spot | 2% heavy air

Bayview delivers the most sheltered profile of all nine venues — essentially no heavy-air risk and a high drifter rate (32%). For a major championship, the light wind resource is a limitation in this window. BYC's world-class facilities and deep regatta history are well known, but the wind climatology for mid-September suggests this calendar window is not its strongest.

Grand Traverse Bay — Grand Traverse YC (GTYC)

Mean 6.4 kt | 63% sweet spot | 1% heavy air

The lightest venue in the dataset and the most direction-variable (steadiness R = 0.04 — essentially no persistent dominant direction). The West Arm of Grand Traverse Bay is surrounded by terrain that disrupts consistent breeze, and 37% of racing hours fall below 5 kt — the highest drifter rate of any candidate. The enclosed bay limits sea breeze development compared to open Great Lakes venues. Grand Traverse Bay is a beautiful venue, but mid-September is not the optimal window for reliable championship racing based on 11 years of HRRR data.

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4. Comparative Rankings by Criterion

Criterion	1st	2nd	3rd	Worst
Mean speed	LYC / MYC (tied, 8.9)	TBYC (8.4)	HYC (8.3)	GTYC (6.4)
Lowest drifter rate (< 5 kt)	MYC (18%)	TBYC (19%)	LYC (21%)	GTYC (37%)
Highest sweet-spot % (5–15 kt)	TBYC (77%)	MYC (73%)	CATYC (72%)	GTYC (63%)
Lowest heavy-air risk (≥ 15 kt)	GTYC (1%)	BYC (2%)	CATYC / NCYC (3%)	LYC (12%)
Capped maximum (predictable ceiling)	TBYC (17.3 kt)	GTYC (16.4 kt)	CATYC (17.8 kt)	LYC (22.6 kt)

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5. Key Takeaways

1. **Lake Michigan venues lead on raw wind resource** — Ludington and Muskegon share the top mean speed (8.9 kt) by a margin. Open lake fetch and consistent sea-breeze development give both a reliable afternoon wind engine.
2. **Muskegon edges Ludington for overall racing quality** — identical speed profiles but a better distribution (73% vs. 67% in sweet spot, 9% vs. 12% heavy air). For a broad fleet, Muskegon is the stronger Lake Michigan option.
3. **Tawas Bay is the outlier worth serious consideration** — 77% sweet spot and only 5% heavy air is the best combined distribution in the dataset. The lower maximum (17.3 kt) means conditions are more predictable, which simplifies race committee decisions and is friendly to a wide range of boat types.
4. **All venues show variable wind direction** — Steadiness R ranges from 0.04 to 0.27. No venue offers a "locked-in" steady direction in mid-September. Race committee experience with shifting conditions is important regardless of venue selection.
5. **The Lake St. Clair venues (BYC, CSYC) are the lightest** — suitable for certain fleet types but carry a 30–32% drifter rate risk for mid-September. Their infrastructure and logistics advantages are real, but the wind climate is a meaningful limitation.
6. **Grand Traverse Bay is not recommended for this window** — 37% drifter rate and R=0.04 (near-zero directional coherence) create significant schedule uncertainty. A different calendar window (July, early August) would likely show a substantially different picture.
7. **Lake Erie offers a balanced middle ground** — Catawba Island and North Cape both land in the 7–7.5 kt range with 70–72% sweet spot and very low heavy-air exposure. If Great Lakes logistics favor Ohio/SE Michigan, these are credible options.

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6. Methodology & Data Notes

Data source: NOAA High-Resolution Rapid Refresh (HRRR) model, 3 km grid, analysis fields (forecast hour 0), retrieved via the Herbie Python library from the NOAA/AWS public archive.

Sample point: Each venue's sample point is offset offshore from the club coordinates along a bearing pointing toward open water (2–4 nm depending on harbor geometry), then spatially averaged over a ±4 nm grid-cell box. This avoids HRRR land-surface cells and approximates on-water conditions.

Wind variables: 10-meter U and V wind components converted to speed (knots) and meteorological direction. Gusts not included in this analysis.

Analysis window: September 8–14, each year, 14:00–20:00 UTC (10 AM–4 PM EDT). Years 2015–2025 (11 years; 2014 excluded because Sep 8 precedes HRRR archive availability of Sep 30, 2014).

Observations: 7 hours/day × 7 days × 11 years = 539 targeted samples per venue; actual counts are ~523 due to occasional missing archive files in early years (2015). HYC/Caseville returned 476 observations — 6 files missing from 2015 — and is slightly underrepresented.

Important caveats: - HRRR is a model, not observed data. It captures synoptic and mesoscale patterns well but may underestimate or smooth localized effects (bay geometry, shoreline topography, thermal triggers). - A single 7-day calendar window analyzed across years captures inter-annual variability but not full seasonal breadth. Results reflect *this specific week* in mid-September, not all of September. - Venue infrastructure, race course geometry, safety logistics, and fleet preferences are outside the scope of this analysis and should be weighted alongside the wind climatology.

Analysis performed using HRRR_club_wind_climatology_v1.0.py. Wind rose and statistics figures for each venue available separately.